

1 Online Learning for Multiclass Classification with Strategic Withholding

In this section, we extend the offline learning framework for multiclass classification with strategic feature withholding to the online learning setting. We derive an online learning algorithm analogous to the regression setting, dealing with the unique challenges of classification, specifically the discrete nature of default classes and the non-convexity of the 0-1 loss.

1.1 Problem Formulation

Consider an online game played over T rounds between a receiver and a sequence of senders. Let $\mathcal{X} \subset \mathbb{R}^d$ be the feature space and $\mathcal{Y} = \mathcal{U} = \{1, \dots, K\}$ be the label and sender preference spaces, respectively.

At each round $t = 1, \dots, T$:

1. The receiver commits to a default class $i_t \in \{1, \dots, K\}$ and a multiclass linear predictor parameterized by $\Theta_t = [\theta_{t,1}, \dots, \theta_{t,K}] \in \mathbb{R}^{d \times K}$.
2. Nature draws a triplet $(u_t, x_t, y_t) \sim \mathcal{P}$, which is observed by the sender.
3. The sender decides to reveal or withhold the features x_t . The revelation indicator is deterministic based on the default class: $A_t = \mathbf{1}\{u_t \neq i_t\}$. The receiver observes the message $M_t = x_t$ if $A_t = 1$ and $M_t = \emptyset$ if $A_t = 0$.
4. The receiver predicts $\hat{y}_t = i_t$ if $M_t = \emptyset$, and $\hat{y}_t = \arg \max_{c \in \{1, \dots, K\}} \langle \theta_{t,c}, x_t \rangle$ if $M_t = x_t$.
5. The receiver incurs the 0-1 loss $\ell_t(i_t, \Theta_t) = \mathbf{1}\{\hat{y}_t \neq y_t\}$.

The overall strategic risk at round t for a fixed pair (i, Θ) is:

$$R_t(i, \Theta) = \mathbf{1}\{u_t = i\} \mathbf{1}\{y_t \neq i\} + \mathbf{1}\{u_t \neq i\} \mathbf{1}\{h_\Theta(x_t) \neq y_t\} \quad (1)$$

where $h_\Theta(x_t) = \arg \max_c \langle \theta_c, x_t \rangle$.

1.2 Surrogate Loss and Bandit Feedback

The 0-1 loss is discontinuous and non-convex. Furthermore, the receiver only observes bandit feedback regarding the default class i_t : the revelation indicator A_t for other defaults $j \neq i_t$ remains unknown.

Following the approach in the online regression setting, we replace the 0-1 loss with a surrogate loss. Specifically, we employ the multiclass logistic loss (cross-entropy) for the predictive component:

$$\phi(\Theta; x, y) = -\log \left(\frac{\exp(\langle \theta_y, x \rangle)}{\sum_{c=1}^K \exp(\langle \theta_c, x \rangle)} \right) = -\langle \theta_y, x \rangle + \log \sum_{c=1}^K \exp(\langle \theta_c, x \rangle) \quad (2)$$

The surrogate strategic loss is thus defined as:

$$\tilde{\ell}_t(i, \Theta) = \mathbf{1}\{u_t = i\} \mathbf{1}\{y_t \neq i\} + \mathbf{1}\{u_t \neq i\} \phi(\Theta; x_t, y_t) \quad (3)$$

Because the loss is non-convex globally (due to the discrete choice of i and strategic interactions), we design an algorithm that achieves local regret or regret bounded within a local basin, similarly to the smooth regression setting. To handle the discrete default classes, we utilize the EXP3 algorithm over the K classes, coupled with Online Gradient Descent (OGD) for the continuous parameter Θ .

1.3 Algorithm

We propose Algorithm 1, which maintains an independent OGD instance for each default class, and uses exponential weights to select the active default class at each round.

Algorithm 1 Online Multiclass Classification with Strategic Withholding

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1: Initialize: For each  $i \in \{1, \dots, K\}$ , initialize  $\Theta_{1,i} \in \mathcal{W}$  and weight  $w_{1,i} = 1$ . Let  $\eta > 0$  be the learning rate for OGD, and  $\gamma \in (0, 1]$  be the exploration parameter for EXP3.  
2: for  $t = 1, \dots, T$  do  
3:   Compute probabilities  $p_{t,i} = (1 - \gamma) \frac{w_{t,i}}{\sum_{j=1}^K w_{t,j}} + \frac{\gamma}{K}$ .  
4:   Sample default class  $i_t \sim p_t$ .  
5:   Commit to default class  $i_t$  and predictor  $\Theta_{t,i_t}$ .  
6:   Receive message  $M_t \in \{x_t, \emptyset\}$ .  
7:   if  $M_t = \emptyset$  then  
8:     Suffer loss  $L_t = \mathbf{1}\{y_t \neq i_t\}$  (if  $y_t$  is revealed) or use an unbiased proxy.  
9:      $\Theta_{t+1,i_t} = \Theta_{t,i_t}$  (No feature update).  
10:  else  
11:    Predict  $\hat{y}_t = \arg \max_c \langle \theta_{t,i_t,c}, x_t \rangle$ .  
12:    Observe true label  $y_t$ .  
13:    Suffer surrogate loss  $L_t = \phi(\Theta_{t,i_t}; x_t, y_t)$ .  
14:    Compute unbiased gradient:  $\hat{g}_t = \frac{1}{p_{t,i_t}} \nabla \phi(\Theta_{t,i_t}; x_t, y_t)$ .  
15:    Update predictor:  $\Theta_{t+1,i_t} = \Pi_{\mathcal{W}}(\Theta_{t,i_t} - \eta \hat{g}_t)$ .  
16:  end if  
17:  For  $j \neq i_t$ , keep  $\Theta_{t+1,j} = \Theta_{t,j}$ .  
18:  Compute unbiased loss estimate:  $\hat{L}_{t,i} = \frac{L_t}{p_{t,i_t}} \mathbf{1}\{i = i_t\}$ .  
19:  Update weights:  $w_{t+1,i} = w_{t,i} \exp\left(-\frac{\gamma}{K} \hat{L}_{t,i}\right)$  for all  $i \in \{1, \dots, K\}$ .  
20: end for
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1.4 Regret Analysis and Proofs

Due to the non-convexity of the true 0-1 risk, we state the regret with respect to the best joint parameters (i^*, Θ^*) over the surrogate loss sequence. We assume the adversary (nature and senders) is oblivious, meaning the sequences (u_t, x_t, y_t) are fixed in advance.

The total regret can be decomposed into two parts: the regret of the EXP3 meta-algorithm for choosing the default classes, and the regret of the OGD base-algorithms for learning the continuous predictors.

Lemma 1 (EXP3 Regret). *Let $L_{t,i} = \tilde{\ell}_t(i, \Theta_{t,i}) \in [0, B]$ be the loss of arm i at round t . Under Algorithm 1, the expected regret of the default class*

selection against any fixed i^* is bounded by:

$$\mathbb{E} \left[\sum_{t=1}^T \tilde{\ell}_t(i_t, \Theta_{t,i_t}) \right] - \mathbb{E} \left[\sum_{t=1}^T \tilde{\ell}_t(i^*, \Theta_{t,i^*}) \right] \leq B\gamma T + \frac{BK \ln K}{\gamma}. \quad (4)$$

Proof. By the standard analysis of the EXP3 algorithm, the importance-weighted loss estimates $\hat{L}_{t,i}$ are unbiased: $\mathbb{E}_{i_t}[\hat{L}_{t,i}] = L_{t,i}$. The exponential weights update guarantees that for any i^* ,

$$\sum_{t=1}^T \mathbb{E}[L_{t,i_t}] - \sum_{t=1}^T \mathbb{E}[L_{t,i^*}] \leq \gamma \sum_{t=1}^T \mathbb{E}[L_{t,i^*}] + \frac{K \ln K}{\gamma} \leq B\gamma T + \frac{BK \ln K}{\gamma}.$$

□

Next, we bound the regret of the continuous predictor updates. Notice that the predictor Θ_{t,i^*} is only updated when $i_t = i^*$. By using importance-weighted gradients, we ensure the OGD algorithm makes progress in expectation over all rounds.

Lemma 2 (OGD with Importance Weighted Gradients). *Let the surrogate loss ϕ be G -Lipschitz with respect to Θ , and let $\mathcal{W}$ have diameter D . For any fixed default i^* and predictor $\Theta^* \in \mathcal{W}$, the OGD update in Algorithm 1 guarantees:*

$$\mathbb{E} \left[\sum_{t=1}^T \tilde{\ell}_t(i^*, \Theta_{t,i^*}) \right] - \sum_{t=1}^T \tilde{\ell}_t(i^*, \Theta^*) \leq \frac{D^2}{2\eta} + \frac{\eta G^2 K T}{2\gamma}. \quad (5)$$

Proof. Let $\hat{g}_{t,i^*} = \frac{\mathbf{1}\{i_t=i^*\}}{p_{t,i^*}} \nabla_{\Theta} \tilde{\ell}_t(i^*, \Theta_{t,i^*})$. Note that $\nabla_{\Theta} \tilde{\ell}_t = \mathbf{1}\{i_t \neq i^*\} \nabla \phi$. Thus $\hat{g}_{t,i^*}$ is an unbiased estimate of the true gradient: $\mathbb{E}_{i_t}[\hat{g}_{t,i^*}] = \nabla_{\Theta} \tilde{\ell}_t(i^*, \Theta_{t,i^*})$. By the standard OGD analysis on the strongly convex Euclidean projection:

$$\langle \hat{g}_{t,i^*}, \Theta_{t,i^*} - \Theta^* \rangle \leq \frac{1}{2\eta} (\|\Theta_{t,i^*} - \Theta^*\|^2 - \|\Theta_{t+1,i^*} - \Theta^*\|^2) + \frac{\eta}{2} \|\hat{g}_{t,i^*}\|^2.$$

Taking expectations and summing over T , the telescoping sum yields $\frac{D^2}{2\eta}$. For the gradient norm, we use the fact that $\|\nabla \phi\| \leq G$ and $\mathbf{1}\{i_t = i^*\}^2 = \mathbf{1}\{i_t = i^*\}$:

$$\mathbb{E}[\|\hat{g}_{t,i^*}\|^2] = \mathbb{E} \left[\frac{\mathbf{1}\{i_t = i^*\}}{p_{t,i^*}^2} \|\nabla \tilde{\ell}_t\|^2 \right] \leq \mathbb{E} \left[\frac{1}{p_{t,i^*}} G^2 \right].$$

Since EXP3 ensures $p_{t,i^*} \geq \frac{\gamma}{K}$, we have $\mathbb{E}[\|\hat{g}_{t,i^*}\|^2] \leq \frac{K}{\gamma} G^2$. Summing over T rounds yields the bound. By the convexity of $\tilde{\ell}_t$ in Θ , $\tilde{\ell}_t(i^*, \Theta_{t,i^*}) - \tilde{\ell}_t(i^*, \Theta^*) \leq \langle \nabla \tilde{\ell}_t(i^*, \Theta_{t,i^*}), \Theta_{t,i^*} - \Theta^* \rangle$. Taking expectations completes the proof. $\square$

We now combine these lemmas to state the final regret bound.

Theorem 1 (Surrogate Regret Bound). *By setting $\gamma = \min\{1, (\frac{K \ln K}{T})^{1/3}\}$ and $\eta = \frac{D}{G} \sqrt{\frac{\gamma}{KT}}$, Algorithm 1 achieves the following expected surrogate regret against any optimal fixed joint parameter (i^*, Θ^*) :*

$$\mathbb{E} \left[\sum_{t=1}^T \tilde{\ell}_t(i_t, \Theta_{t,i_t}) \right] - \min_{i, \Theta} \sum_{t=1}^T \tilde{\ell}_t(i, \Theta) \leq \mathcal{O} \left(T^{2/3} (K \ln K)^{1/3} + DGT^{2/3} K^{1/3} \right) = \tilde{\mathcal{O}}(T^{2/3}). \quad (6)$$

Proof. By summing the bounds from Lemma 1 and Lemma 2, the total regret is bounded by:

$$\text{Regret} \leq B\gamma T + \frac{BK \ln K}{\gamma} + \frac{D^2}{2\eta} + \frac{\eta G^2 KT}{2\gamma}.$$

Substituting $\eta = \frac{D}{G} \sqrt{\frac{\gamma}{KT}}$, the OGD regret terms balance to $DG \sqrt{\frac{KT}{\gamma}}$. The total regret becomes $B\gamma T + \frac{BK \ln K}{\gamma} + DG \sqrt{\frac{KT}{\gamma}}$. Optimizing over γ , we find that $\gamma \propto T^{-1/3}$ yields a total regret of $\tilde{\mathcal{O}}(T^{2/3})$. $\square$

The above result highlights the fundamental difference between the bandit regression and classification settings. While regression with one-point continuous bandit feedback typically incurs an $\mathcal{O}(\sqrt{T})$ or $\mathcal{O}(T^{3/4})$ regret depending on the assumptions, the classification setting restricts the bandit problem purely to the discrete choice of default class, naturally leading to a $\tilde{\mathcal{O}}(T^{2/3})$ regret bound via the interaction between EXP3 exploration and unbiased OGD updates.